FIRECAN

An interactive map for filtering, Displaying & exporting historical forest fires across Canada

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github.com/thomascheung05/FIRECAN

Why FIRECAN?

The problem

- Canadian fire perimeters are public but scattered across agencies, formats and projections
- Quebec publishes its own data, split into pre- and post-1976
- CWFIS covers all of Canada but lacks very old fires
- Every clean subset (“fires > 500 ha within 50 km of X since 1990”) means GIS wrangling - Barrier to information

The goal

- One merged dataset, one map, point-and-click filters, easy export

Data sources

Region	Source
Quebec	Données Québec – *Feux de forêt*
Rest of Canada	CWFIS – NFDB
Watersheds	Esri Canada

Common schema: `fire_year`, `fire_size` (ha), `province`, `data_source`, `geometry` (EPSG:4326)

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How it works

- **Backend:** Python · Flask · GeoPandas · Shapely (`app/firecan_main.py`, `app/firecan_fx.py`)
- **Frontend:** single-page HTML + Leaflet (`static/`)
- **Caching:** processed data stored as GeoParquet; first run pulls pre-built files from GitHub Releases (~2 GB), falling back to raw downloads

Logic flow

Data setup

- Checks if data exists locally
- If not, downloads from GitHub or the raw source
- Data is processed and cached

Request loop

- User picks filters in the frontend
- Filters are applied to the data
- Results are sent back for display and export

What you can do with it

Filters (combined with AND)

- Province (one, several, or all)
- Year range (min / max)
- Fire size in hectares (min / max)
- Radius around a lat/lon point (buffered in a local UTM CRS, so km are real km)
- Within a selected watershed (via the Watershed Explorer)

Performance knob

- Polygon tolerance → `simplify()` geometries to trade detail for load time

Export filtered set

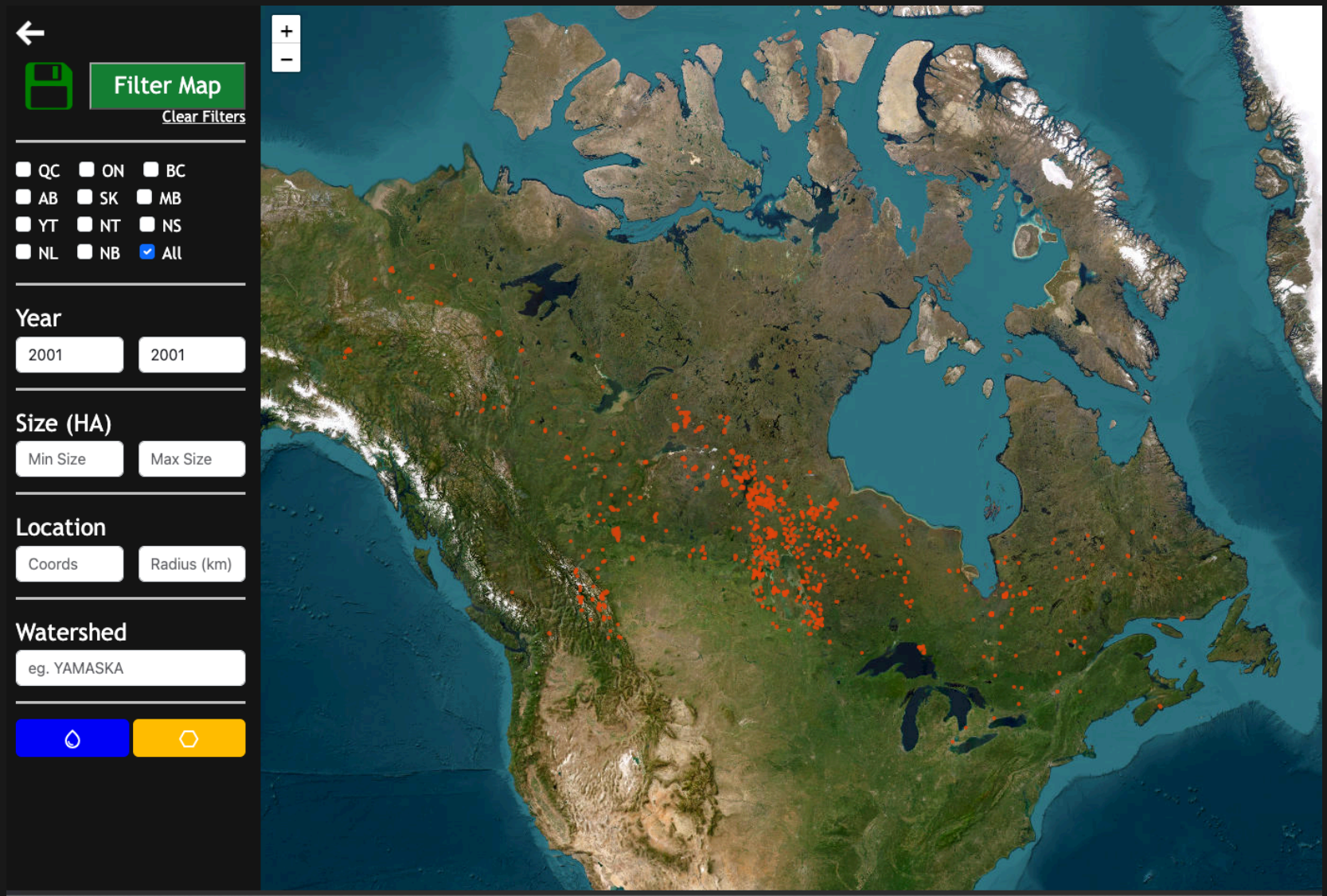
- CSV
- GeoJSON
- GeoPackage

Getting started

```
git clone https://github.com/thomascheung05/FIRECAN
cd FIRECAN
./FIRECAN.sh    # builds venv, installs deps,
                # fetches data, opens browser
```

Runs locally at `http://127.0.0.1:5000`

Filtering All Of Canada



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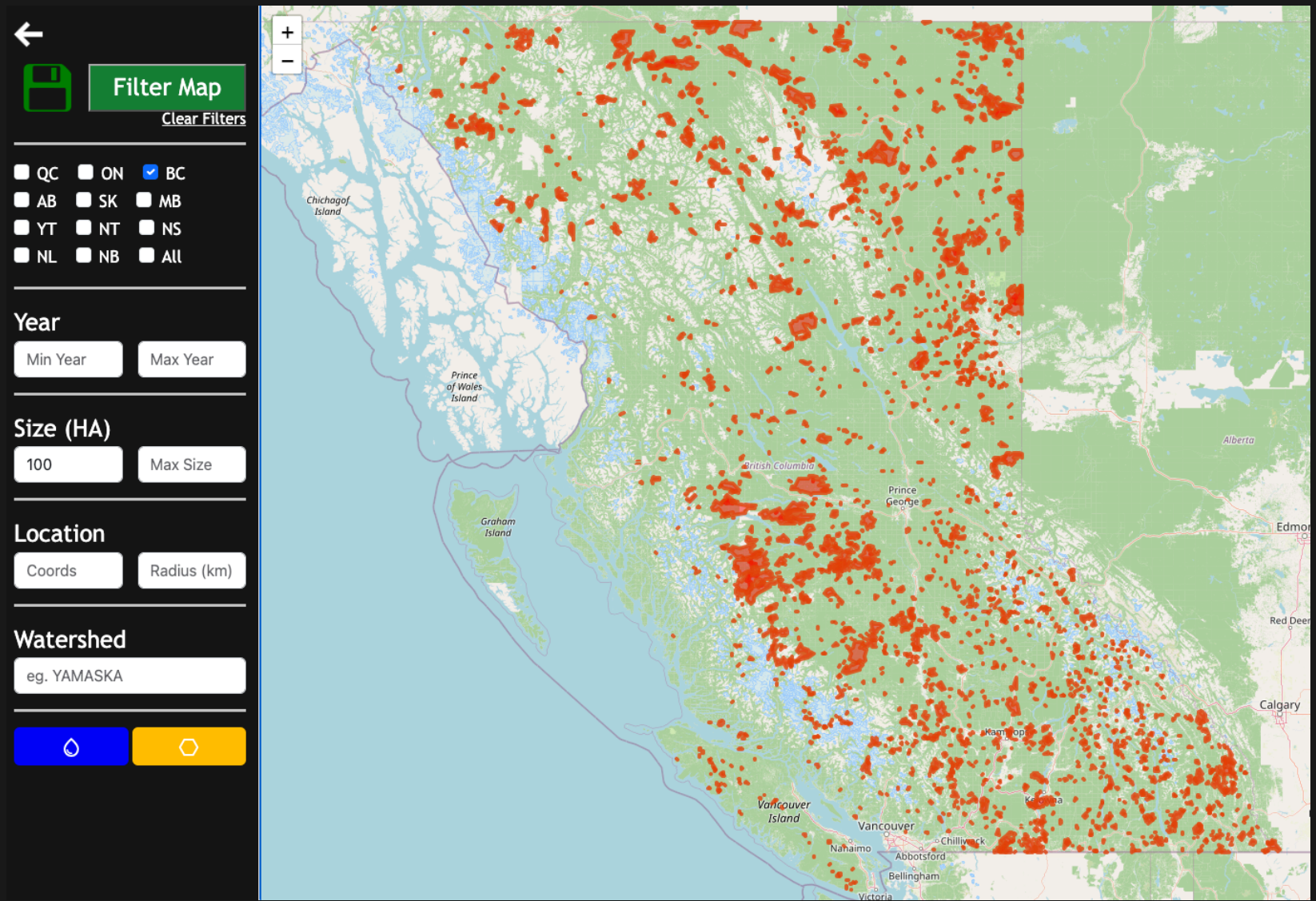
Filtering BC



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Filtering Watershed

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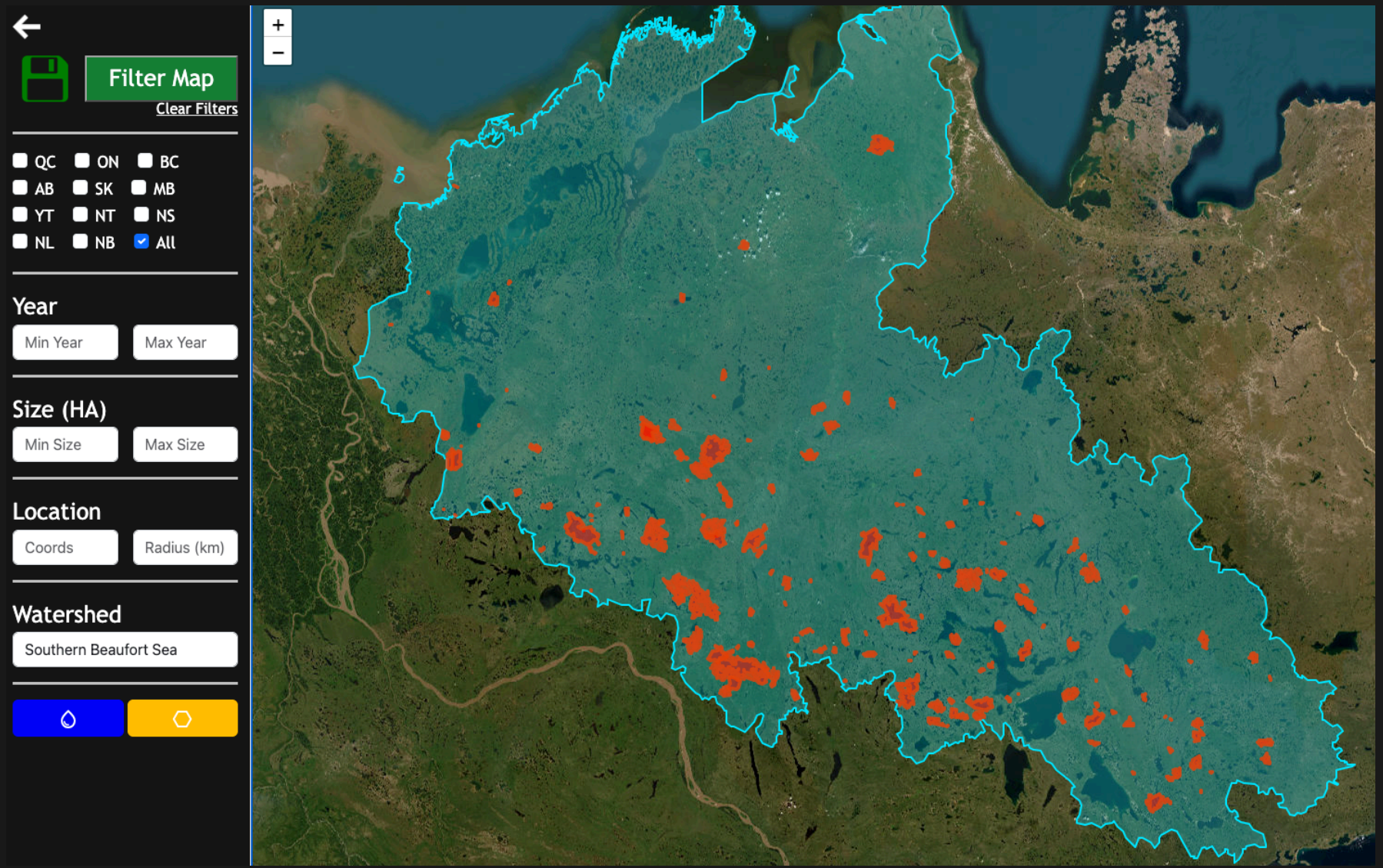
Filtering Watershed



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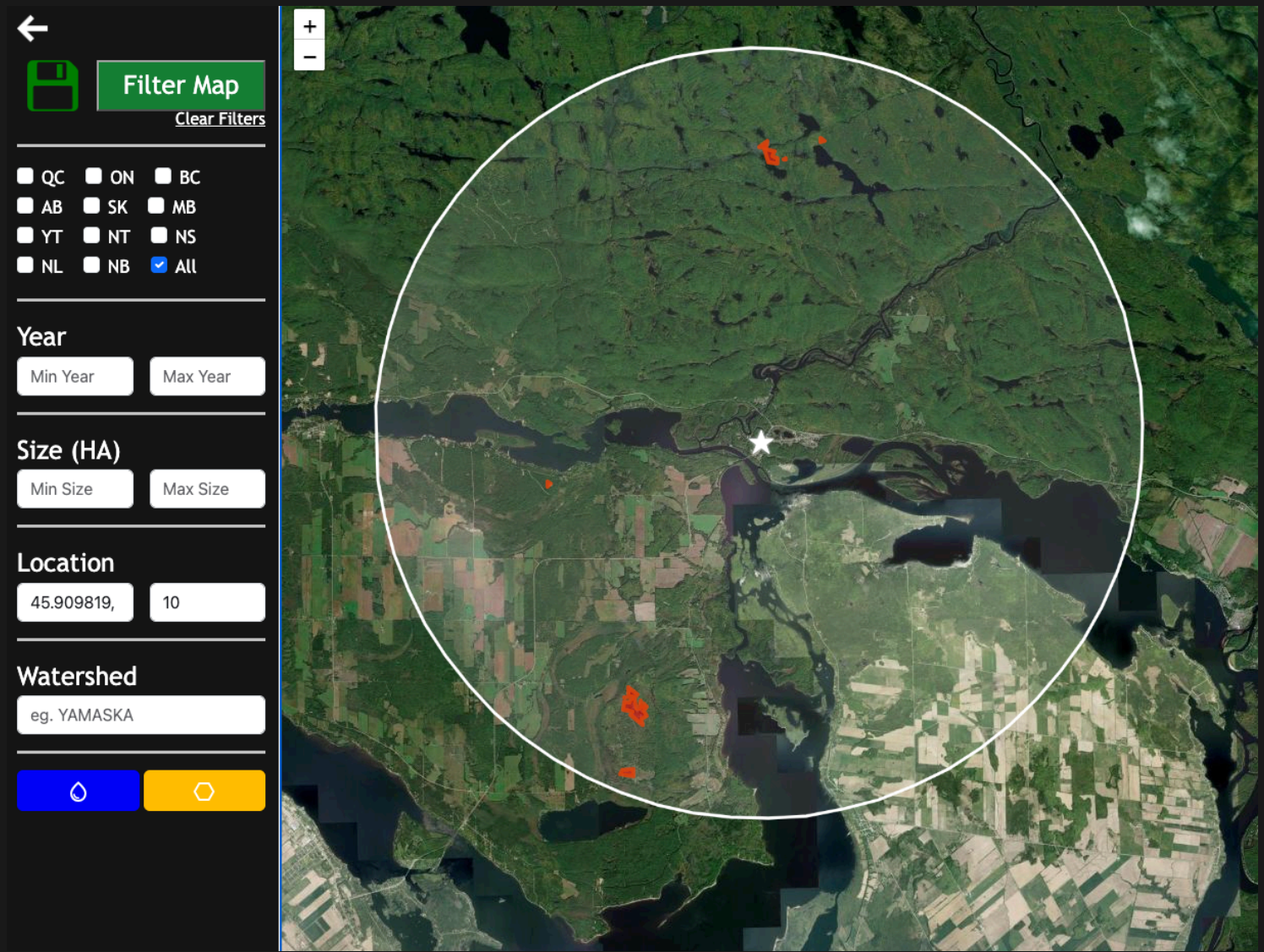
Filtering Radius



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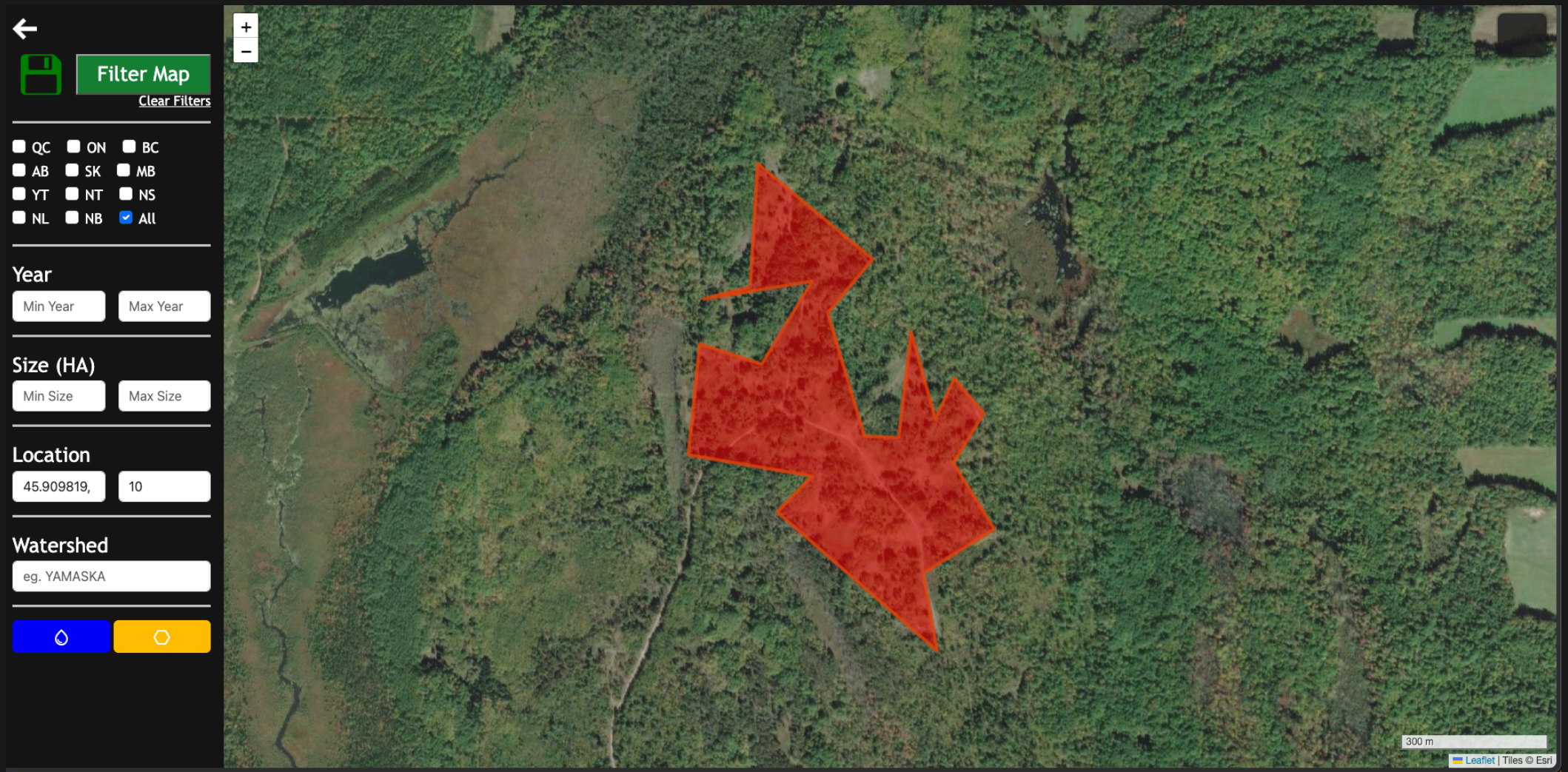
Polygon Tolerance



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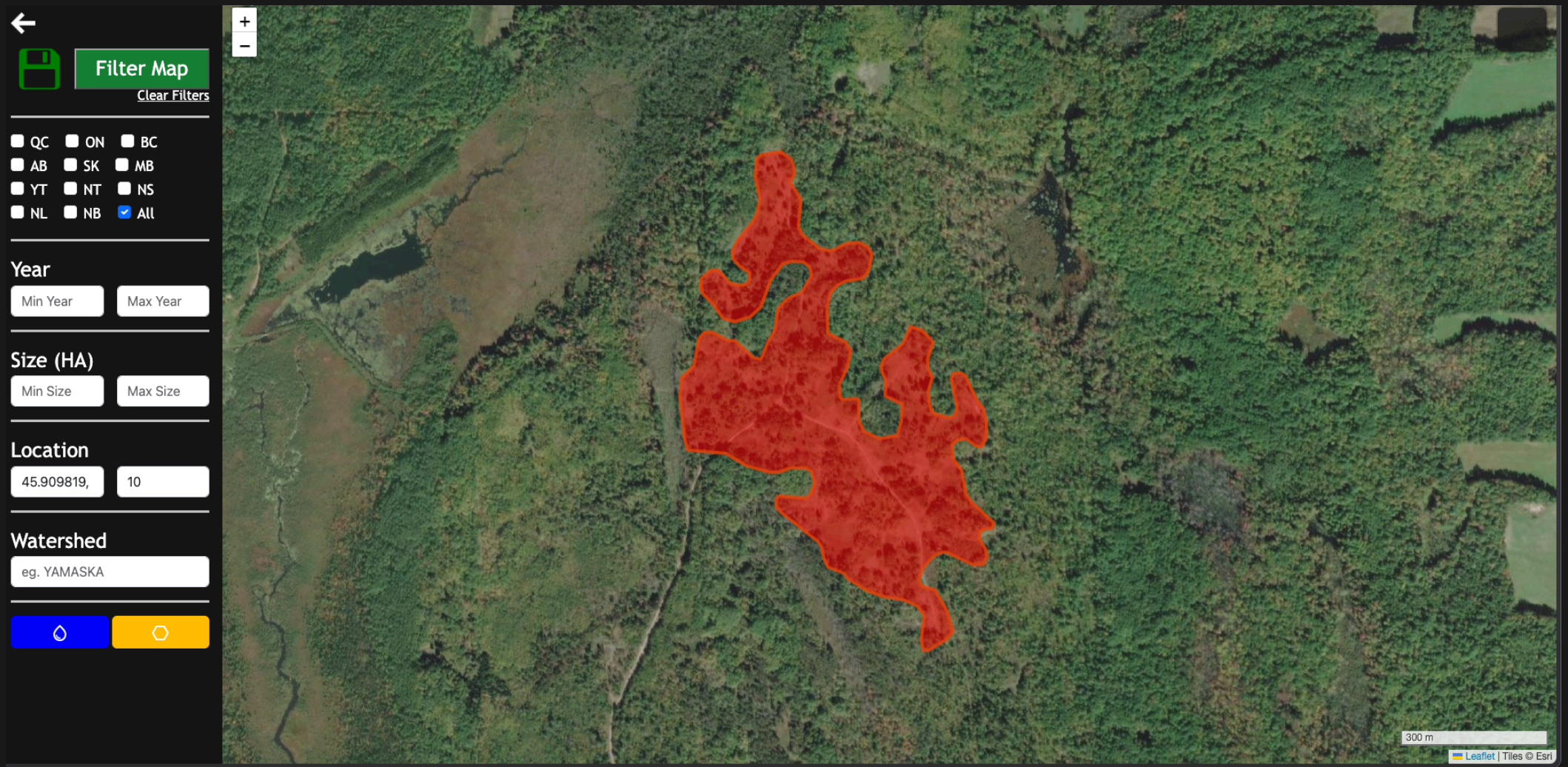
Polygon Tolerance



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Status, next steps & feedback

Current limitations

- Datasets are large (efficiency)
- Runs locally only (no hosted instance yet)
- Limited Exposure
- Not tested in practical setting

Planned

- Onboarding Brad Henry
- Let users **upload their own polygon** (parks, study areas, tenure blocks…) and filter by it
- Update UI
- Get and apply feedback

Feedback

- What features would be beneficial to your work
- How familiar are you with running python apps
- How useful would it be to host it online
- Keep it Canadian or expand
- Other outputs (maps, summary tables)
- UI quick fixes

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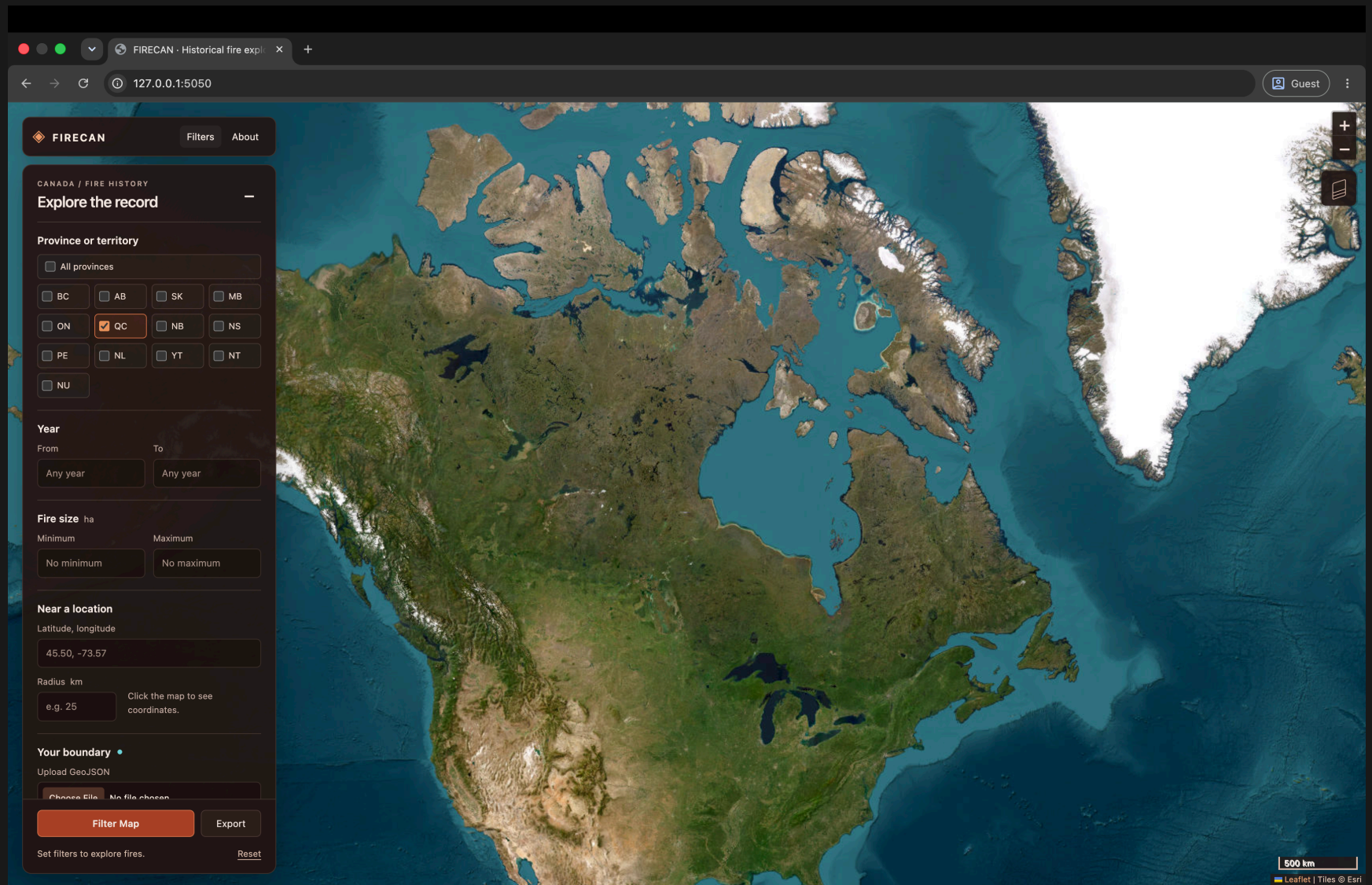
New UI



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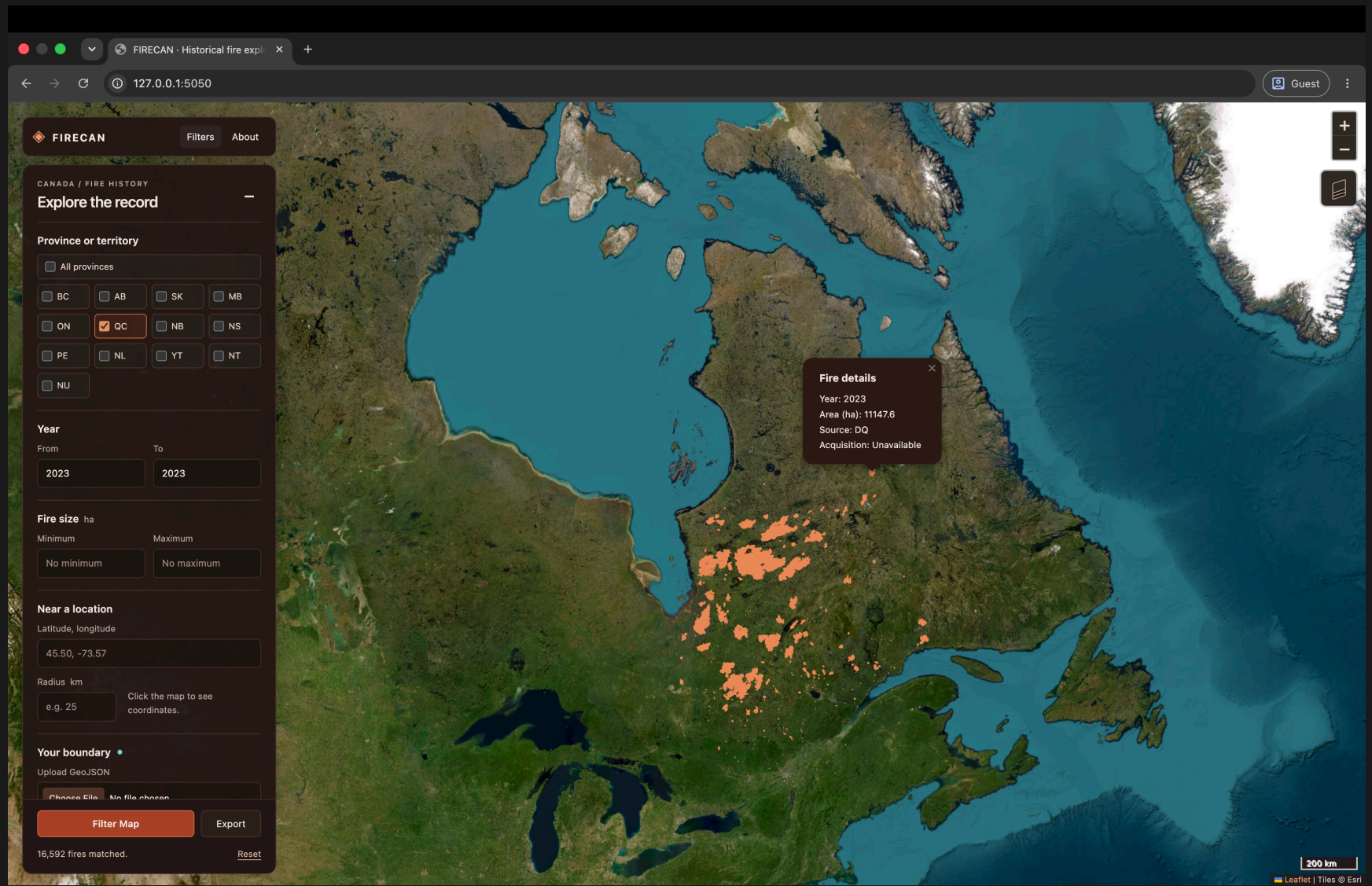

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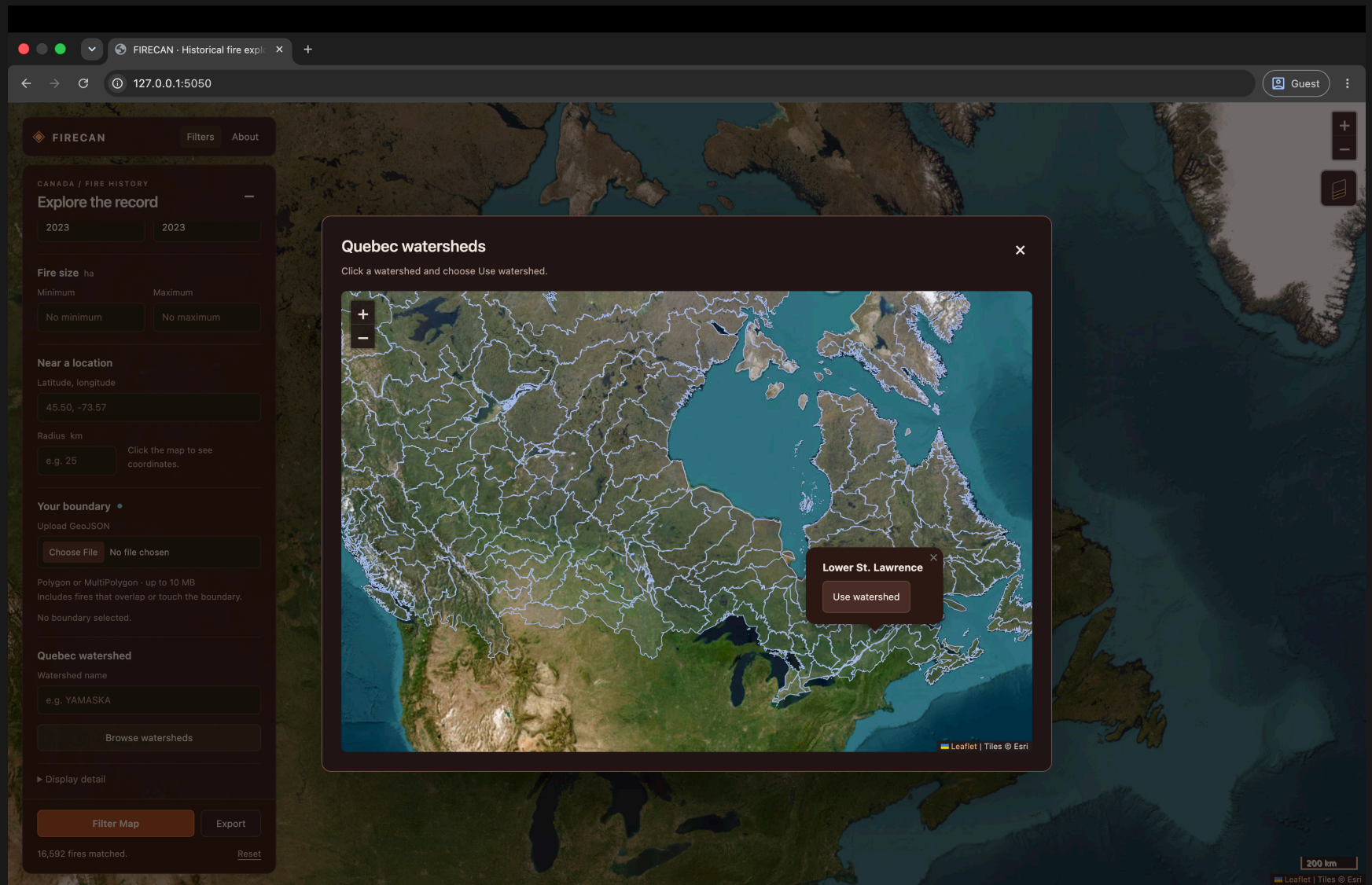

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